

VLADIVOSTOK, Russian Federation

WMO: 319600

Lat: 43.1201N

Lon: 131.9155E

Elev: 189

StdP: 99.08

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-24.1	-21.3	-30.2	0.2	-21.1	-28.0	0.3	-19.4	14.8	-12.8	13.2	-14.3	2.0	0	0.677

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.8	28.1	21.3	26.2	20.5	24.6	19.9	23.2	25.8	22.3	24.4	21.4	23.3	3.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.4	17.4	24.1	21.6	16.6	23.4	20.8	15.8	22.6	70.1	25.6	66.6	24.2	63.4	23.3	27.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.0	11.3	10.0	-26.8	31.2	4.9	2.2	-30.4	32.7	-33.2	34.0	-36.0	35.2	-39.6	36.8		
(5)			-27.3	25.0	4.8	1.2	-30.8	25.9	-33.5	26.6	-36.2	27.2	-39.7	28.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.3	-12.9	-8.9	-1.5	6.0	10.9	15.1	19.3	21.0	16.4	8.8	-1.6	-10.1	(6)
(7)		DBStd	12.13	4.53	4.82	4.43	3.46	3.28	3.20	2.98	2.33	3.01	4.24	5.72	4.91	(7)
(8)		HDD10.0	2798	710	530	356	127	28	2	0	0	1	73	348	624	(8)
(9)		HDD18.3	4926	968	763	614	369	231	106	24	4	72	295	597	883	(9)
(10)		CDD10.0	1077	0	0	0	8	56	154	290	340	192	36	1	0	(10)
(11)		CDD18.3	162	0	0	0	0	0	8	55	86	13	0	0	0	(11)
(12)		CDH23.3	658	0	0	0	2	8	67	242	296	43	1	0	0	(12)
(13)		CDH26.7	131	0	0	0	1	1	12	58	56	5	0	0	0	(13)
(14)	Wind	WSAvg	4.3	4.8	4.4	4.6	4.7	4.5	4.1	3.9	3.8	3.8	4.4	4.6	4.4	(14)
(15)	Precipitation	PrecAvg	809	12	17	25	50	76	96	140	156	116	66	36	20	(15)
(16)		PrecMax	1261	64	66	93	150	187	280	403	306	352	234	133	80	(16)
(17)		PrecMin	452	0	0	0	6	4	9	20	4	6	1	0	0	(17)
(18)		PrecStd	204	13	16	19	31	47	61	84	72	79	47	31	19	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	1.4	5.3	12.0	21.4	24.2	28.2	30.9	30.4	27.1	20.2	13.8	5.2	(19)
(20)			MCWB	-0.7	0.8	5.0	11.2	14.7	19.9	22.8	22.1	19.2	14.0	9.1	3.8	(20)
(21)		2%	DB	-1.2	2.1	8.5	17.1	21.2	25.1	28.1	28.1	24.2	18.1	10.9	2.3	(21)
(22)			MCWB	-2.8	-0.6	2.9	9.4	13.0	18.1	21.8	21.5	17.9	13.3	7.5	0.5	(22)
(23)		5%	DB	-3.2	0.6	6.3	14.0	18.9	22.9	26.0	26.2	22.5	16.6	9.0	0.1	(23)
(24)			MCWB	-4.9	-1.4	1.7	7.3	12.0	17.0	21.1	21.1	17.5	12.8	6.4	-1.7	(24)
(25)		10%	DB	-5.2	-1.0	4.5	11.8	16.8	20.5	24.0	24.8	21.0	15.2	7.0	-2.0	(25)
(26)			MCWB	-7.1	-2.9	0.8	6.1	11.1	16.1	20.4	20.7	16.7	12.0	4.4	-3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	1.8	6.0	12.1	16.1	21.2	24.5	24.4	21.6	16.7	10.6	4.3	(27)
(28)			MCDWB	0.9	3.4	10.2	19.7	22.6	26.0	28.1	27.5	24.1	18.4	11.9	5.0	(28)
(29)		2%	WB	-2.6	0.2	3.7	10.0	14.3	19.4	23.2	23.4	20.0	15.4	8.5	1.1	(29)
(30)			MCDWB	-1.6	1.4	6.8	15.9	19.4	23.5	26.6	25.7	22.3	16.7	9.9	1.9	(30)
(31)		5%	WB	-4.8	-1.2	2.5	8.3	13.1	18.0	22.0	22.6	19.0	14.1	6.7	-1.6	(31)
(32)			MCDWB	-3.2	0.4	5.3	12.4	17.1	21.2	24.7	24.4	20.8	15.7	8.6	0.0	(32)
(33)		10%	WB	-7.2	-3.0	1.4	6.8	12.0	16.8	21.0	21.9	18.2	12.9	4.7	-3.8	(33)
(34)			MCDWB	-5.3	-0.9	3.8	10.2	15.4	19.5	23.2	23.5	20.0	14.4	6.8	-2.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.4	9.9	8.5	9.1	9.2	7.9	6.8	8.8	9.3	8.7	9.1	(35)	
(36)			MCDBR	12.7	12.7	11.7	14.6	14.4	13.3	11.5	10.4	11.8	11.7	12.1	12.4	(36)
(37)			MCWBR	11.3	10.3	7.5	7.9	7.6	7.4	6.2	5.4	6.6	7.9	9.5	11.1	(37)
(38)		5% WB	MCDBR	12.0	11.3	10.7	13.8	12.2	11.7	10.0	7.7	8.5	8.6	10.9	11.5	(38)
(39)			MCWBR	11.3	9.5	7.4	8.0	7.3	7.0	5.9	4.6	5.5	6.9	9.4	10.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.376	0.454	0.546	0.570	0.549	0.519	0.472	0.421	0.430	0.382	0.335	(40)	
(41)		taud	2.194	2.030	1.895	1.742	1.763	1.912	2.064	2.198	2.256	2.080	2.116	2.145	(41)	
(42)		Ebn at Noon	788	790	769	730	727	745	764	789	802	723	697	721	(42)	
(43)		Edh at Noon	94	132	173	218	219	190	161	137	120	125	103	91	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.19	3.15	4.24	4.70	4.96	4.92	4.47	4.49	4.23	3.16	2.18	1.81	(44)	
(45)		RadStd	0.14	0.20	0.28	0.42	0.47	0.71	0.59	0.42	0.38	0.23	0.18	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	(e) +2.32	(f) N/A	(g) -1.45	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) -88	(m) -44
(47)	Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page